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Sooner or Later? Health Information Technology, Length of Stay, and Readmission Risk

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Adequate length of stay (LOS) during hospitalization is not only a critical determinant of quality of care, but can be a useful predictor of the risk of future readmissions. Recent studies have shown alarming evidence that the United States leads developed nations in terms of shorter hospital stays, rendering patients with greater risk of future readmissions. We focus on deviation between hospital LOS and the geometric mean LOS (GMLOS), a guideline for care delivery stipulated by the Centers for Medicare and Medicaid Services (CMS). Our objective is to establish the relationship between LOS and readmission risk, examine the role of health IT in reducing the deviation of LOS, and address cost trade-offs between early discharges and readmissions. Based on a large panel of congestive heart failure (CHF) patients during a 4-year period, we find that implementation of health IT applications is associated with a reduction in the deviation between LOS and GMLOS. This deviation is associated with a significant increase in future 30-day readmission risk. Patients whose inpatient stays are shorter than the GMLOS guideline, by 1 day or more, are likely to exhibit a 1.1% greater risk of readmission, and even shorter stays are likely to further exacerbate their readmission risk. Further, we find that the total readmission costs are much higher than the costs incurred in keeping patients longer during their initial hospital visit. Our results have important policy implications for providers and hospital administrators with respect to their discharge policies of CHF patients.

Key words: length of stay; congestive heart failure; readmission risk; health information technology
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1. Introduction

Effectively managing the length of patient stay is a critical operational decision for most hospitals. Patient length of stay (LOS) has been widely considered as an operational measure to assess operations efficiency (Glick et al. 2003, McDermott and Stock 2007), quality of care (Angst et al. 2011, Hwang et al. 2002), and resource allocation decisions of hospitals (Rappaport et al. 2003). LOS is a key component of healthcare cost that resonates across the healthcare system, including patients, healthcare providers, and payers (e.g., insurers and CMS). Reducing excessive LOS has attracted much attention and focus of policymakers in the Affordable Care Act (ACA).[1] The $3 million Heritage Health Prize Competition highlights the significance of LOS as a critical component of the current healthcare system. The competition, which was announced in 2011, provided an opportunity for participants to develop the best performing analytics models to predict LOS for the members of Heritage Providers Network based on 3 years of de-identified, patient data. Hence, understanding the determinants and the consequences of LOS are of paramount importance for both academic researchers and healthcare practitioners.

In this study, we explore the antecedents of LOS from the lens of health IT (HIT), and the consequence of LOS on patients' readmission risk. In the extant literature, the use of HIT has been primarily linked to healthcare outcomes such as readmission and mortality risk. Angst et al. (2011) conclude that strategic implementation of HIT induces higher quality in terms of the Average Length of Stay (ALOS).

[Correction added on 10 September 2018, after first online publication: the affiliation of Indranil R. Bardhan has been changed.]

However, despite its importance, the role of HIT and its impact on variations in LOS is an under-researched topic. In this research, we explore the role of HIT from a new perspective. Specifically, we investigate the link between HIT and the deviation of LOS from the standard Geometric Mean of Length of Stay.[2] Our focal interest is on the extent of the deviation of LOS from GMLOS, because (a) complying with GMLOS is of particular financial interest to healthcare providers since it is the basis for reimbursement by CMS, and (b) GMLOS provides a guideline for standardized care based on diagnosis related groups (DRGs). Hence, by investigating the relationship between HIT usage and deviations of LOS from GMLOS, we explore their impact on hospitals' adherence to standard LOS guidelines stipulated by CMS.

We focus on the impact of LOS deviation on the quality of care delivery as measured by the 30-day, patient readmission rate, that is, the likelihood of a patient to be readmitted (as an inpatient) for the same principal diagnosis, within 30 days of discharge from the previous visit (Halfon et al. 2006). The 30-day readmission risk has become a central barometer of the ACA to assess the quality of care delivery (Helm et al. 2016). Starting in 2012, CMS has imposed penalties on hospitals for preventable readmissions related to chronic conditions such as heart failure or pneumonia (Bardhan et al. 2015). In this study, we are interested in studying the extent to which deviations from the GMLOS guidelines (as mandated by CMS) impact patient readmission risk.

Early discharge decisions may increase short-term benefits by reducing short-term costs associated with patient visits and increase hospital throughput. However, if patients are discharged prematurely before receiving sufficient care, their risk of readmission may increase. Therefore, the initial short-term cost savings may be offset by the cost of additional readmissions in the future. It is our objective to quantify this cost trade-off between early discharge and readmissions.

Hence, our study addresses the following questions: (a) what is the association between implementation of health IT and hospital operational efficiency, measured in terms of their compliance with standardized LOS guidelines (i.e., GMLOS)?; (b) what is the impact (if any) of deviations from these LOS guidelines on future readmission risk?; and (c) what are the cost trade-offs between LOS deviation and future readmissions?

Our findings indicate that HIT implementation enables hospitals to comply with standard LOS practice guidelines. Through a difference-in-difference (DID) estimation approach, we observe that operational HIT applications reduce the overall LOS deviation from diagnostic-related standard guidelines at both the patient discharge level and the hospital level. We find that shorter LOS (i.e., lower than GMLOS) increases patient readmission risk. While discharging a patient earlier can save healthcare expenditures in the short term, our analysis indicates that it may come at the expense of extra care incurred in future readmissions, which tend to be costlier.

Our research makes several key contributions. First, we investigate the impact of operational health IT on hospitals' adherence to best practice guidelines. Unlike previous studies which predominantly adopted LOS as the primary outcome measure, our study represents the first attempt at measuring hospital operational efficiency based on deviations from the GMLOS guidelines. Second, we capture the trade-offs between the short- and long-term effects of critical operational decisions by jointly examining the role of discharge decisions and readmission risks associated with patient admissions. Third, our study contributes to the growing health policy literature by quantifying the economic impact of early discharges on the overall, long-term cost of care delivery and coordination. Our study also has implications for hospital planners with respect to capacity and resource planning, since early discharge of patients may have beneficial short-term impact but negative long-term consequences on staff and bed planning decisions.

2. Literature Review

2.1. Health IT and Length of Stay

The length of stay associated with an inpatient (hospital) visit has been considered to be a significant indicator of the quality of care delivery and a measure of the resources expended in hospital operations (Anderson et al. 2014, Andritsos and Tang 2014, Kc and Terwiesch 2012). While the impact of HIT on the cost and quality of healthcare delivery has been extensively studied in the prior literature (e.g., Brenner et al. 2016), few studies have addressed the role of HIT on patient length of stay, a significant factor in assessing the performance and quality of care of healthcare providers. Angst et al. (2011) and Forgionne and Kohli (1996) suggest that HIT implementation is associated with reductions in LOS; Lee et al. (2015) develop a decision support system with a machine learning framework and report improved timeliness of care and shortened LOS. However, these studies were conducted at the hospital level where they studied the impact of HIT on the average LOS within a hospital. On the other hand, our analysis is conducted at the level of an individual patient admission, enabling us to control for admission-specific factors, including a variety of patient-specific factors. Another distinguishing feature of our study is that we examine the deviation from GMLOS for each admission, rather than the absolute value of LOS associated with a patient visit. In other words, we identify the factors impacting deviations from LOS guidelines in order to subsequently study their impact on future readmission risk, a topic which has not been studied in the prior literature.

2.2. Length of Stay and Readmission Risk

A review of the literature indicates that the association between reductions in LOS and healthcare outcomes is mixed. Since the introduction of the prospective payment system in 1983, evidence suggests that patients are leaving hospitals quicker and sicker (Kahn et al. 1990, Kosecoff et al. 1990). For example, the average hospital stay for surgical cases in the United States has dropped from 11 days in 1981, to 8.5 days in 1991; and from 9.4 days to 7 days for nonsurgical cases (Baker et al. 2004). High workload and overwork can account for premature discharge of cardiac patients and may be associated with higher mortality rates (Kc and Terwiesch 2009). On the other hand, analysis of patient admission data from a single general internal medicine unit finds that increasing LOS does not have a significant impact on the likelihood of readmissions (Keenan et al. 2008).

At a macro level, recent studies have presented alarming evidence that the United States leads most developed nations in terms of shorter hospital stays, but is at greater risk of future readmission due to complications (Joynt and Jha 2013). A national study of patients reported that about 15% of all CHF patients were discharged in an unstable condition, with confusion or incontinence, which exposed them to a higher mortality risk (Kosecoff et al. 1990). From a clinical perspective, adequate LOS during a patient hospitalization is necessary to ensure full recovery and mitigate future readmission risk. On the other hand, excessive LOS leads to greater resource expenditures.

Adequate LOS during hospitalization is not only a determinant of the current quality of care delivery; it is also critical in reducing the likelihood of future patient visits (Straney et al. 2012). Prior studies have documented a declining trend in the average LOS of Medicare patients who are hospitalized for heart failures (Bueno et al. 2010), for example, from 8.8 days to 6.3 days from 1993 to 2006 for CHF patients (Keenan et al. 2008). These studies also report that, between 1993 and 2006, in-hospital mortality decreased from 8.5% to 4.3%, while 30-day mortality rate decreased by 2.1%. These findings suggest that shorter LOS of a hospital visit does not necessarily compromise quality of care on the current visit. However, a finer analysis conducted during this period reported that 30-day hospital readmission rates increased from 17.2% to 20.1%, while patient discharges to skilled nursing facilities increased from 13% to 19.9%. It stands to reason that shorter LOS trades off against higher future readmission risk as a result of relinquishing care during patient hospitalization and greater reliance on post-acute care after discharge.

There is a small but growing interest in studying the relationship between LOS and readmissions. Andritsos and Tang (2014) examined patients' resource usage measured by the total LOS incurred due to unplanned readmissions within 30 days after initial inpatient hospitalization. Kc and Terwiesch (2012) found that patients, who are discharged early from the cardiac intensive care unit (ICU) due to high occupancy, had a higher risk of being readmitted and, hence, consumed additional future resources.

Nevertheless, prior research does not systematically study the joint effects of HIT, LOS and readmission risk; nor does it prescribe an appropriate benchmark LOS for patient visits. In this research, we examine the trade-offs between LOS and readmission risk and identify LOS deviation parameters or safety corridors for CHF inpatients. Our benchmark, GMLOS, is determined from Medicare data, which provides evidence-based guidelines on the suggested LOS. The GMLOS represents the national mean length of stay for each diagnosis related group (DRG), as determined by CMS.

In sum, our research differs from prior studies in several respects. First, we specifically examine the role of health IT on adherence to healthcare standards, measured in terms of deviation from GMLOS, which has not been investigated in the prior literature. Second, we consider the trade-off between deviations from GMLOS and the forward-looking readmission risk of a patient's visit. Third, from a methodological perspective, we apply difference-in-differences estimation and propensity score matching to analyze the effect of health IT implementation on the deviation of LOS and patient readmission risk, to account for the potential endogeneity issues related to the impact of HIT on readmission risk.

2.3. Health IT and Readmission Risk

The impact of health IT on healthcare quality has been extensively studied in the prior literature. A majority of recent studies document a positive impact of health IT on hospital outcomes, including hospital quality (effectiveness), efficiency of care, patient safety, and satisfaction (Buntin et al. 2011). A recent systematic review by Jones et al. (2014) also reports that there is strong evidence to support the use of clinical decision support and computerized provider order entry (CPOE) systems to improve the quality, safety, and efficiency of health care.

However, recent studies on the impact of health IT on patient-level health outcomes have yielded mixed evidence. For example, Bardhan and Thouin (2013) and Adler-Milstein et al. (2015) report that EHR adoption is associated with better hospital performance, measured in terms of greater adherence to process quality and increased patient satisfaction. DesRoches et al. (2010) report that adoption of electronic health records (EHR) does not significantly change 30-day readmission rates, for diagnoses such as acute myocardial infarction (AMI) and congestive heart failure (CHF). Based on hospital-level data, McCullough et al. (2013) do not find a significant relationship between health IT usage, readmission rates, and hospital length of stay. However, the use of EHRs may be indirectly associated with better quality of care. Amarasingham et al. (2010) report that a real-time electronic predictive model, using data extracted from EMRs, leads to higher accuracy in identifying heart failure patients at greater risk for readmission. Based on analysis of CHF patient data across 6 years, Bardhan et al. (2015) conclude that the use of cardiology information systems is associated with lower readmission risk, while administrative IT system usage is associated with lower frequency of future readmissions.

3. Research Data

In this section, we first describe our data collection process followed by a description of the variables used in our model development and analysis.

3.1. Data Collection

Our data consist of patient admission records from 67 non-Federal hospitals in the North Texas region, starting from January 2006 to December 2009. Hence, a patient visit constitutes our unit of analysis. We focus on patients with CHF as their primary diagnosis, since it represents one of the four major, chronic diseases under the purview of the Hospital Readmissions Reduction Program (HRRP), established in 2012. Patient readmissions across multiple hospitals are tracked by matching the regional entity master patient index (REMPI), developed by the Dallas Fort Worth Hospital Council Research Foundation. The REMPI allows us to obtain the readmission history of patients *across multiple hospitals* of different health systems. This is a significant improvement over previous studies which have been restricted to the use of readmission data from hospitals that belong to a single system, with a limited scope due to small sample size or focus on a specific class of patients such as Medicare beneficiaries (Silverstein et al. 2008).

Our data consist of 65,210 patient admissions from 40,989 distinct patients with CHF as the primary diagnosis. Among these patients, 70% had a single admission, while 30% experienced multiple admissions during this period. Within the 65,210 admissions, 24,221 cases are readmissions among which 7029 cases represent 30-day readmissions. Table 1 tabulates descriptive statistics on our model variables, which includes patient demographics, visit characteristics, payer type, admission condition, discharge destination, and other hospital characteristics. While 51% of the discharged patients are female, 67% of all discharges are Caucasian, and 25% are African-American patients. With respect to 30-day readmissions, 46% are female patients, 61% are of Caucasian race, and 33% are African Americans. The average discharge age is 69 years, with 66% of the patients being 65 years or older. About 62% of all admissions belong to Medicare patients. We observe that the average LOS is 5.45 days, with an average of 12.58 diagnoses and 1.08 procedures, and an average charge of $37,649 per visit. For 30-day readmissions, the average LOS is 6.3 days, with an average of 1.08 procedures, and an average charge of $41,966 per visit. Almost 47% of the patients exhibit a moderate risk of mortality of score of 2 (on a scale of 1 to 4), while 38% of the patients have mortality risk score of 3 or higher. For readmitted patients, 45% have moderate mortality risk score of 2, while 43% of the patients have higher risk mortality score.

3.2. Variable Description

Our data include admission characteristics such as LOS, number of procedures, payer type, admission type, and the risk of mortality. LOS is defined as the number of days of patient stay from admission to discharge in an inpatient setting. Since each discharge reports the diagnosis related group (DRG) code, we measure the deviation in length of stay from GMLOS, by first matching the DRG code of an individual patient discharge and the corresponding DRG-specific GMLOS, based on a given year. We then calculate the deviation between the reported LOS and GMLOS.

Payer type has been used in the literature as a covariate to study patient readmission rates (Amarasingham et al. 2010, Bradley et al. 2013). We classify *Payer type* into five categories based on the patient's insurance claim filing code: Medicare, Medicaid, self-pay, private insurance, and other types (veteran's plans or other federal programs). For each admission, a hospital records the patient's risk of mortality. Patient mortality risk is coded on a scale from 1 to 4, which represents the death risk of a patient as minor, moderate, major, or extreme, respectively.

Upon readmission, a patient can return to the same hospital as the prior admission, or choose a different hospital. Our sample statistics indicate that 37% of readmitted CHF patients visited a different hospital compared to their prior visit. For each readmission, we count the number of different hospitals visited by

Table 1 Variables and Descriptive Statistics

Variable name	Description of variable	All admissions[a]	30-day readmissions only[a]
Demographic variables			
Gender	Patient's gender	Female 51%; Male 49%	Female 46%, Male 54%
Race	Patient's race	White (67%), African American (25%)	White (61%), African American (33%)
Disch_Age	Patient's age at the day of discharge	69 (15.65)	67 (16.15)
Visit characteristics			
LOS	Actual length of patient stay	5.45 (5.87)	6.3 (7.26)
LOS − GMLOS	Geometric Mean LOS − Actual LOS	−1.13 (5.75)	1.87 (6.85)
Procedures	Number of procedures performed	1.08 (1.98)	1.08 (2.08)
Pt_stickiness	Number of pt. visits to same hospital divided by all visits until current admission	0.93 (0.18)	0.78 (0.27)
Prior_Admission	No. of prior pt. admissions at current visit	0.89 (1.92)	2.97 (3.18)
Insurance type			
Medicare	Binary indicator of Medicare insurance	40,282 (61.77%)	4,117 (58.57%)
Medicaid	Binary indicator of Medicaid insurance	12,256 (18.79%)	1,650 (23.47%)
Private	Binary indicator of Private insurance	41,650 (63.87%)	4,080 (58.05%)
Other	Binary indicator of other types of insurance (VA, workers comp, etc.)	686 (1.05%)	77 (1.10%)
Admission condition			
Risk Mortality	Patient risk of death (1: Minor; 2: Moderate; 3: Major; 4: Extreme)	1: 9,869 (15.13%), 2: 30,568 (46.88%), 3: 19,070 (29.24%), 4: 5,703 (8.75%)	1: 901 (15.13%), 2: 3,158 (44.93%), 3: 2,297 (32.68%), 4: 673 (9.57%)
Discharge destination facility			
Intermediate Care	Discharge to intermediate care facility	1,213 (1.86%)	139 (1.98%)
Short Term	Discharge to another short-term general hospital for inpatient care	1,191 (1.83%)	167 (2.38%)
Home	Discharge to home health services org.	8,174 (12.53%)	958 (13.63%)
Skilled Nursing	Discharge to skilled nursing facility	7,109 (10.90%)	817 (11.62%)
Long-term Care	Discharge to long-term care hospital	1,286 (1.97%)	170 (2.42%)
Others	Other types of facilities	5,651 (8.67%)	971 (13.81%)
Hospital characteristics			
Beds	Number of hospital beds	392.70 (298.44)	
Urban	Binary indicator of urban/rural hospital's area (1 = urban/0 = rural)	68,597 (4,768)	
Teaching	Binary indicator of teaching hospital	46.5% (0.002)	
CMI	Case Mix Index	1.54 (0.27)	

[a]Standard errors are shown in parentheses unless represented otherwise.

the patient prior to their current visit. We introduce a variable, *Patient Stickiness*, computed as the number of patient visits to the same hospital divided by the total number of visits, up to the current visit. Since clinicians do not always have electronic access to a patients' prior medical history, the quality of care delivery may be affected by the lack of information sharing across hospitals especially when patients migrate across hospitals to obtain care (Ayabakan et al. 2017).

Other hospital characteristics may also affect patient readmission rates. We include a number of hospital-level control variables that have been used in the prior literature including hospital *size* measured in terms of the number of beds (Friedman et al. 2008, McGarry et al. 2016), hospital case mix index (CMI) (Au et al. 2012, Flood et al. 2013), geographical location, and teaching status (Friedman et al. 2008, McGarry et al. 2016). *Beds* represent the number of beds available at a hospital, a measure of hospital capacity. Hospital CMI[3] accounts for the average severity of patient case mix in a hospital. *Location* codes the geographical location of a hospital and is classified as urban or rural, based on hospital zip codes. It takes a value of one if the hospital is located in an urban area, and zero otherwise. Hospital size and teaching status are considered to be possible factors which may impact readmission risk since teaching hospitals and larger hospitals (in general) have better capability in treating more severe patients (Theokary and Ren 2011). On the other hand, larger hospitals may be more cost-efficient and effective in implementing safety programs and patient counseling programs at discharge, thus reducing the risk of readmission (Friedman et al. 2008).

We also include the patient discharge destination, which indicates the type of the care facility to which a patient is discharged. Using patient discharges to home care as the reference baseline, other patient discharge destinations include intermediate care facility, short-term general hospital for inpatient care, home health services organization, skilled nursing facility, long-term care hospital, and other types of facilities. Discharge destination is a possible risk factor for readmission since it reflects the severity of a patient's condition, as well as an indicator of post-acute care coordination (McGarry et al. 2016, Robbins and Webb 2006).

3.3. Health Information Technology

Hospital health IT usage data are drawn from the HIMSS Analytics database for the corresponding 4-year period, that is, from 2006 to 2009. We select HIT applications that are associated with patient care coordination, and that may directly affect their length of stay. The list of such HIT applications includes Cardiology Information System, Operating Room IS for surgery (pre-operative), Enterprise Resource Planning, Clinical Decision Support, CPOE, Order Entry (includes order communications), Enterprise EMR, RFID for patient tracking, Business Intelligence, Chart Tracking/Locator, Laboratory Information System, Nurse Staffing/Scheduling, Electronic Medication Administration Record (EMAR), Patient Scheduling, OR Scheduling, and Bed Management systems. Detailed descriptions of each application are presented in Table 2.

Nurse Staffing/Scheduling, for example, is an application that coordinates scheduling of nurses and staffs for inpatient services and identifies conflicts with other appointments for patients. These systems enable care inpatient providers to improve their operations efficiency by reducing unforeseen resource conflicts, reducing time spent in finding alternatives, rescheduling appointments, and providing better care coordination during patient visits (Schimmelpfeng et al. 2012). Consequently, they can help shorten the length of stay of patients without compromising care quality. Similarly, *OR Scheduling* systems support all functions necessary to schedule patients in the operating room, including scheduling surgeons, materials, equipment, and surgical suites for surgical services. With OR Scheduling, a patient may need less time waiting for surgery or may receive required treatments in a timelier manner, and hence, experience a shorter hospital LOS. Hence, we posit that these types of health IT can supplement providers with accurate patient information and decision support which enables compliance with GMLOS guidelines.

4. Econometric Models

In order to estimate the impact of HIT on the deviation of LOS from GMLOS, we adopt a difference-in-difference (DID) estimation method and a propensity score matching approach.

4.1. Difference-in-Difference Estimation

A standard DID analysis creates a quasi-experimental setup by comparing a treatment group and a control group, before and after treatment. DID methods have been widely used in empirical economics since Ashenfelter (1978) and Ashenfelter and Card (1985). This method is usually associated with "natural experiments" when there are policy changes that can be used to effectively define treatment and control groups, and hence, obtain an unbiased treatment effect (Imbens and Wooldridge, 2009).

A DID analysis first examines whether a statistically significant difference exists for a subject before and after treatment (i.e., the first difference); then it identifies a baseline for comparison between the treatment and the control group (i.e., the second difference). By comparing the change (before and after treatment) in the treatment group relative to the control group, DID enables researchers to control for (unobserved) heterogeneity across groups (Goldfarb and Tucker 2014, Rishika et al. 2013). There are two underlying assumptions to ensure consistent DID estimation. First, it is assumed that the time effects are common across treated and untreated individuals. Second, if cross-section data are used, the composition of the treated and untreated groups is assumed to be stable before and after the change (Cameron and Trivedi 2005, p. 770).

In our case, the first difference in the DID estimation is the change in the deviation of LOS, pre- and post-implementation of health IT, for a patient visit to a specific hospital. Our HIMSS data allow us to observe the implementation year for each HIT application at each hospital. DID mimics a quasi-experiment situation and its estimation addresses several other sources of endogeneity that may result from selection bias and omitted variables, not captured in the dataset. The DID method is designed such that any bias caused by variables (observed or unobserved) common to the treated and control groups are implicitly controlled or washed away (Goldfarb and Tucker 2014). In our case, the second difference in the DID estimation is the change in the deviation of LOS among patient visits to hospitals with HIT implementation (treatment) relative to hospitals without HIT (control). Therefore, the estimated DID effects can be attributed to the treatment effect, that is, implementation of HIT applications.

Following this approach, we first calculate the deviation of LOS among hospitals, before and after each HIT application was implemented. We code the HIT application as a binary variable where one indicates that it has been implemented and operational, and zero indicates otherwise.[4] Therefore, we can track the change in an application's status turning from inactive to fully operational. Observing such a marker for

Table 2 Health IT Application Definitions

Health IT application	Percentage of hospitals with full impln. (year 2009)	Definition
Bed Management	41%	An application that is used to manage the use of beds in the hospital. This application tracks when beds are available and creates notifications to cleaning staff to prepare the bed for the next patient, and then notifies registration/administration when the bed/room has been cleaned and is available for the next patient
Business Intelligence	51%	This software explores operational key performance indicators that enable organizations to cut costs and make decisions in real-time thus improving bottom line results
Cardiology Information System	63%	An application that specifically automates functions in the cardiology department. The application must provide some of the following: order processing, permanent patient history index maintenance, image and EKG tracing storage, transcribing and distributing results, clinical documentation, prep instruction cards maintenance, appointment scheduling, and management reporting
Chart Tracking/Locator	86%	Chart Location/History function allows the updating and review of the location of a patient's medical records chart. It helps to track Medical Record Charts with Check-out, Check-in, and reporting capabilities
Clinical Decision Support	83%	An application that uses pre-established rules and guidelines, that can be created and edited by the healthcare organization, and integrates clinical data from several sources to generate alerts and treatment suggestions
Computerized Practitioner Order Entry (CPOE)	33%	An order entry application specifically designed to assist practitioners in creating and managing medical orders for patient services or medications. This application has special electronic signature, workflow, and rules engine functions that reduce or eliminate medical errors associated with physician ordering processes
Electronic Medication Administration Record (EMAR)	54%	eMAR is an electronic record keeping system that documents when medications are given to a patient during a hospital stay. This application supports the five rights of medication administration (right patient, right medication, right dose, right time, and the right route of administration)
Enterprise EMR	36%	Electronic Medical Record (Enterprise EMR represents an environment where the same EMR application is used for inpatient and ambulatory care) ambulatory care
Enterprise Resource Planning	24%	A business management system that integrates multiple business applications including human resources, payroll, materials management, supply procurement, accounts payable, general ledger, and supply chain management by providing an automated and integrated view of business information and reports of data from several operational areas of the business. This system is a tool used to improve the accuracy of resource management and the reduction of operating costs
Laboratory Information System	86%	An application to streamline the process management of the laboratory for basic clinical services such as hematology and chemistry. This application may provide general functional support for microbiology reporting, but does not generally support blood bank functions. Provides an automatic interface to laboratory analytical instruments to transfer verified results to nurse stations, chart carts, and remote physician offices. The module allows the user to receive orders from any designated location, process the order and report results, and maintain technical, statistical, and account information. It eliminates tedious paperwork, calculations, and written documentation while allowing for easy retrieval of data and statistics
Nurse Staffing/Scheduling	67%	An application that automates decisions about staffing, nursing stations, and scheduling nurses' time. May include functions that enable a hospital to quickly review and generate its nurse scheduling; adjust staffing and scheduling based on patient volume, acuity, and staff ability; keep records for budgeting; produce management reports on productivity and census; and maintain records on personnel qualifications
Operating Room (Surgery)—Pre-operative	70%	An OR application that provides clinical documentation/management of relevant pre-surgery information and patient preparation for surgery. It also provides for the management of relevant pre-surgery availability/scheduling/reservation/preparation of room, OR supplies/meds, and staff
OR Scheduling	83%	A scheduling application that addresses all functions necessary to schedule patients, all required clinicians, materials, equipment, and surgical suites for surgical services provided by the healthcare organization
Order Entry (Includes Order Communications)	84%	A legacy HIS application that allows for entry of orders from multiple sites including nursing stations, selected ancillary departments, and other service areas; allows viewing of single and composite results for each patient order. This function creates billing records as a by-product of the order entry function
Patient Scheduling	87%	An application that coordinates scheduling of all the provider components for required patient services and identifies conflicts with other appointments for the patients or provider components. It may include preparation requirements, resource and materials requirements, staff workload lists, and patient care notifications
RFID—Patient Tracking	19%	An RFID system that is used to track patient

each application, we compare the before and after effects of HIT implementation using the regression model specified in Equation (1),

$$y_{ijkt} = \alpha_1 HealthIT\ App_{ijk} + \alpha_1 AfterHealthIT\ App_{ijkt} + \beta HealthIT\ App_{ijk} \times AfterHealthIT\ App_{ijkt} + \varepsilon_{ijkt} \quad (1)$$

where y is the outcome, denoting the deviation of LOS from GMLOS, i indexes individual discharge (of a patient's in-hospital visit), j represents the hospital, k denotes an individual HIT application, t is the time (in year), and e_{ijkt} is the error term. The key focus of this specification is on the interaction term β, which captures the DID effect. $HealthITApp_{ijk}$ takes a value of one if patient i visits hospital j where HIT application k has been implemented, while the variable $AfterHealthITApp_{ijkt}$ takes a value of one if patient i's visit to hospital j during time period t occurs during the post-treatment (after implementation) period.[5]

4.2. Propensity Score Matching

The DID approach does not address a potential source of endogeneity, when the assignment of the treatment and the control groups may not be completely random. In other words, endogeneity may become a concern when a non-random assignment is a result of the outcome variable (Bertrand et al. 2004). Though unlikely, it might be argued that a hospital chose to implement certain HIT applications in response to high deviation from GMLOS, our dependent variable. For example, hospitals that are motivated to streamline their LOS may self-select into implementing HIT applications as an organizational effort. Hence, organizational behavior, rather than the use of HIT applications, may account for the change in the deviation of LOS from GMLOS. We then resort to a second approach, propensity score matching, to alleviate this concern.

Propensity score matching has been used to address selection bias in observational studies of causal effects in the healthcare literature (Crown 2014). In a randomized experiment, the randomization of subjects (e.g., patients) to different treatments guarantees that, on average, there should be no systematic differences in observed or unobserved covariates between subjects assigned to the different treatments. However, in observational studies, such control over treatment assignments is not possible, and the treated and non-treated groups may have large differences on their observed covariates. These differences can lead to biased estimates of treatment effects. The propensity score, defined as the conditional probability of being treated (given the covariates), can be used to balance the covariates in the two groups, and therefore reduce this bias through matching (D'Agostino 1998). In our study, we apply the propensity score matching method to address two concerns: (a) endogeneity due to self-selection bias at the hospital level, and (b) sample selection bias at the patient level.

4.2.1. Propensity Score Matching at the Hospital Level. Propensity score matching involves pairing treatment and control units that are otherwise similar in terms of their observable characteristics (Dehejia and Wahba 2002). Here, the treatment group consists of hospitals which implemented an HIT application, whereas the control group consists of hospitals which did not implement HIT. The observable covariates we use in matching hospitals are the number of beds, hospital teaching status, location, and CMI. We calculated the propensity score of a hospital, given these exogenous covariates. Then, each subject in the treated group was matched to the sample in the control group based on the closest propensity score (Rosenbaum and Rubin 1983). Finally, we estimate the differences in the average after-treatment effects of each HIT application to the absolute deviation of LOS from GMLOS.

4.2.2. Propensity Score Matching at the Patient Level. In this study, we are interested in determining whether early discharge has a causal effect on future readmission risk. Hence, the treatment effect focuses on whether a patient hospital visit is associated with early discharge based on the GMLOS guidelines. We compare the treatment and the control groups who are similar in terms of control variables that affect the outcomes, except for the treatment.[6] We use gender, age, and risk mortality as the exogenous variables to match patients. To construct a matched sample, we first calculate the propensity score of being discharged earlier than the GMLOS (treatment), given the exogenous covariates. Then, each subject in the treatment group is matched with corresponding subjects in the control group (i.e., patients discharged at GMLOS), by identifying the control subject with the closest propensity score (Rosenbaum and Rubin 1983). We then estimate the differences in readmission risk between the early discharge group (treatment) and normal discharge group (control).

4.3. Logistic Regression

The readmission literature has extensively used logit regression models to estimate the readmission probability of patients (Kociol et al. 2012, Ross et al. 2008). These studies model a binary outcome in terms of whether a readmission occurs within 30 days of a patient's discharge from the previous visit (Silber et al. 2003). We also adopt a logit regression model to estimate the probability of readmissions, p, as specified in Equation (2),

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \times |LOS_i - GMLOS_i| + \gamma_1 \times X_j + \gamma_2 \times Y_i + \gamma_3 \times Z_j \tag{2}$$

where X_j represents a binary vector of health IT applications. Y_i represents patient-specific control variables, such as the number of prior admissions, patient stickiness, gender, log(discharge age), (log (discharge age))^2, patient race, risk mortality, payer type, median income, and discharge disposition. Z_i represents a vector of hospital-specific control variables, such as capacity measured in terms of the log of number of beds, teaching status, a binary indicator of urban location, and CMI.[7]

The dependent variable represents the incidence of a readmission, which takes a value of one if a patient is readmitted within a 30-day period from the date of previous discharge, and zero otherwise. Patients in our study are classified into two groups based on the magnitude of their deviation from GMLOS. The "Below Geom. Mean" group consists of visits where LOS – GMLOS < −1 days, whereas the "Around Geom. Mean" group consists of patient visits where the actual LOS is within a 1-day range of the GMLOS, that is, −1 ≤ LOS – GMLOS ≤ +1 day.

5. Results

5.1. Difference-in-Difference Estimations of LOS Deviation

We are interested in whether HIT helps hospitals to improve their adherence to CMS-mandated guidelines related to the GMLOS. Therefore, we compare the deviation in LOS from GMLOS (i.e., |LOS − GMLOS|) for patients admitted to hospitals without health IT applications, against the deviation for patients admitted to hospitals after implementation of health IT, that is, recorded in HIMSS as fully operational. Our control group is composed of patient admissions to hospitals without health IT applications, and the treatment group consists of admissions to hospitals with health IT. In Table 3, we present the average "before-and-after" changes in LOS deviation and the estimation results of the coefficient of the interaction term (β) in our DID model in Equation (1).[8] This coefficient represents the difference-in-difference estimate, which measures the implementation effect of HIT on the |LOS − GMLOS| deviation.

As shown in Table 3, the deviation |LOS − GM LOS| decreases significantly after implementation of several health IT applications. The significant coefficients of the interaction term suggest that the treatment effect of implementing health IT applications on the LOS deviation is negative and significant for Cardiology Information System, Clinical Decision, Computerized Practitioner Order Entry (CPOE), Patient Tracking RFID, and Business Intelligence. In other words, we find that implementations of these HIT systems are associated with significant reductions in the deviation of LOS from the standard guidelines. The largest impact is observed for patient tracking RFID system, with an estimated reduction of 2.655 days.

5.2. Logistic Regression and 30-day Readmission Risk

We now report our findings on the impact of the deviation between the actual LOS and GMLOS on 30-day patient readmission risk, as presented in Table 4. The focus of our interest is on the deviation,

Table 3 Difference-in-Difference Estimation of LOS Deviation from GMLOS

Health IT application	LoS dev. before IT app. impl.	LoS dev. after IT app. impl.	Coeff. of interaction term	SE
Cardiology Information System	3.295	3.016	−0.956	0.307***
Operating Room (Surgery)—Pre-operative	2.527	3.216	0.314	0.401
Enterprise Resource Planning	3.49	3.687	0.038	0.308
Clinical Decision Support	3.15	2.83	−0.876	0.440*
Computerized Practitioner Order Entry (CPOE)	3.266	2.97	−0.784	0.326*
Order Entry (Includes Order Communications)	2.804	2.92	−0.216	0.573
RFID—Patient Tracking	3.718	3.041	−2.655	0.539***
Business Intelligence	3.082	2.469	−0.97	0.288***
Chart Tracking/Locator	3.375	3.139	−0.437	0.628
Laboratory Information System	3.278	2.985	−0.519	0.551
Nurse Staffing/Scheduling	3.029	3.259	−0.039	0.474
Electronic Medication Administration Record (EMAR)	3.159	3.214	−0.578	0.420
Patient Scheduling	3.25	2.662	−0.852	0.583
OR Scheduling	3.313	2.897	−0.522	0.411
Bed Management	3.769	2.744	1.073	0.179***

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$; †$p < 0.1$.

Table 4 Logistic Regression Estimation of 30-day Readmission Risk

Variable	Parameter estimate	SE	Odds ratio
LOS Deviation from GMLOS	0.068	0.017***	1.07
HIT Applications			
EMAR	0.091	0.054†	1.095
Patient Scheduling	−0.206	0.103*	0.814
Number of prior admissions	0.191	0.01***	1.21
log (Beds)	0.063	0.038†	1.065
Patient Stickiness	−0.291	0.117*	0.748
Teaching Hosp.	0.123	0.063†	1.131
Urban Hosp.	0.088	0.105	1.092
CMI	−0.511	0.113***	0.6
Gender: Female	−0.233	0.046***	0.792
log(Discharge Age)	1.454	0.857†	4.28
(log(Discharge Age))^2	−0.229	0.109*	0.795
Race			
African American	0.160	0.059**	1.174
Asian/Pacific Islander	0.088	0.216	1.092
Other	−0.092	0.104	0.912
Risk Mortality			
RM2	0.195	0.064**	1.215
RM3	0.329	0.075***	1.39
RM4	−0.115	0.157	0.891
Payer Type			
Medicare	0.055	0.055	1.057
Medicaid	0.128	0.062*	1.137
Private Insurance	−0.032	0.054	0.969
Other	0.326	0.187†	1.385
Socio-economic Variable			
Median Income	0.000	0.001	1
Discharge Destination			
Intermediate Care Facility	0.561	0.152***	1.752
Short-term Care Hospital	1.687	0.111***	5.403
Home Health Service Org.	0.337	0.074***	1.401
Skilled Nursing Facility	0.201	0.099*	1.223
Long-term Care Hospital	0.003	0.285	1.003
Other	0.151	0.089†	1.163
Constant	−4.328	1.725*	0.013
Adjusted R^2	0.064		
Prob > χ^2	0.000		
Classification accuracy	90.15%		

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$; †$p < 0.1$. Only health IT applications, that are significant at $p < 0.1$, are shown due to space constraints.

|LOS − GMLOS|. Our results indicate that the odds of readmission increase by 7% for patients in the "Below Geom. Mean" group (odds ratio = 1.07). Hence, compared to the patient group where the deviation is around the guideline (|LOS − GMLOS| < =1 day), a CHF patient who is discharged early (by more than 1 day) is 7% more likely to be readmitted within 30 days.

Next, we observe that HIT applications, such as *Patient Scheduling* and *EMAR*, are significantly associated with 30-day readmission risk. For example, implementation of patient scheduling applications account for an 18.6% decrease in the odds of 30-day readmission (odds ratio = 0.814). Patient demographics and visit characteristics are also associated with future readmission risk. *Number of prior admissions* is positively correlated with 30-day readmission risk, increasing the odds of readmission by 21%. The odds of 30-day readmission due to *Patient Stickiness* is 25% lower, suggesting that patients who are treated at the same hospital are less likely to be readmitted, compared to those who migrate across hospitals. This result suggests that CHF patients who return to the same hospital are more likely to reduce their likelihood of future readmissions. In addition, compared to Caucasian patients, African-American patients are 17% more likely to be readmitted, while female patients are 21% less likely to be readmitted within 30 days compared to their male counterparts, which is consistent with the findings in prior literature (Amarasingham et al. 2010, Robbins and Webb 2006, Woz et al. 2012). Specifically, Woz et al. (2012) provide a possible explanation that male patients do a poorer job of understanding and following up on appointments, and hence, exhibit higher risk of being readmitted than female patients.

Our results also indicate that Medicaid patients, when discharged earlier than the GMLOS standard, are likely to exhibit greater readmission risk compared to self-pay patients, which is consistent with prior studies (Amarasingham et al. 2010, Bradley et al. 2013, Robbins and Webb 2006). Specifically, the odds of a 30-day readmission for Medicaid patients is 13.7% higher compared to self-pay patients. As noted by Robbins and Webb (2006), this finding reflects the underlying socio-economic factors behind patient insurance status, which may be critical in continuous care management of chronic conditions.

Patient discharge destination is a significant predictor of future readmission risk. Patients who are discharged early to an intermediate care facility are likely to exhibit 75% greater odds of readmission. Patients who are discharged to a short-term care facility fare even worse: they are five times more likely to be readmitted within 30 days, compared to patients who are discharged to their homes. Similarly, the early-discharge patients to home health organizations and skilled nursing facilities are 40% and 22% more likely to be readmitted, respectively. These results are also consistent with prior findings (McGarry et al. 2016). On the other hand, we find that patient discharged to long-term care hospitals are not likely to exhibit higher readmission risk, which is consistent with previous studies (Kahn et al. 2013).

5.3. Propensity Score Matching Results

In Table 5, we present the average treatment effects of HIT applications on LOS deviations, where hospitals with HIT implementation are in the "treated" group. The estimation results indicate that, after matching hospital characteristics, such as bed size, teaching

Table 5 Health IT Average Treatment Effects after Propensity Score Matching

Variable	Application	Treated	Controls	Difference	SE
$\lvert LOS - GMLOS \rvert$	Cardiology Information System	1.645	2.746	−1.101	0.602*
	Operating Room (Surgery)—Pre-operative	1.686	0.903	0.783	0.444†
	Enterprise Resource Planning	1.686	1.084	0.601	0.281*
	Clinical Decision Support	1.569	2.115	−0.547	0.292†
	Computerized Practitioner Order Entry (CPOE)	1.523	0.929	0.594	0.268*
	Order Entry (Includes Order Communications)	1.658	2.642	−0.984	0.451*
	Enterprise EMR	1.772	1.217	0.555	0.254*
	RFID—Patient Tracking	1.470	2.449	−0.979	0.333**
	Business Intelligence	1.571	2.220	−0.649	0.259*
	Chart Tracking/Locator	1.636	2.519	−0.883	0.409*
	Laboratory Information System	1.646	2.784	−1.138	0.601†
	Nurse Staffing/Scheduling	1.660	2.453	−0.793	0.376*
	Electronic Medication Administration Record (EMAR)	1.605	2.544	−0.939	0.375*
	Patient Scheduling	1.640	1.386	0.254	0.883
	OR Scheduling	1.641	3.050	−1.409	0.656*
	Bed Management	1.310	1.887	−0.577	0.345†

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$; †$p < 0.1$.

status, and urban location, most of the HIT applications are significantly associated with reduced deviations of length of stay (as indicated by the negative sign on the difference term). Among the HIT systems, Cardiology information systems, Laboratory information systems, and OR scheduling systems, have the most significant effect on reducing |LOS − GMLOS| deviations by more than a day (on average). For example, Cardiology information systems account for a reduction in the average deviation of LOS (from GMLOS) by 1.1 days, significant at 5%. Hospitals with Laboratory information system exhibit a lower average deviation by 1.1 days, while hospitals with OR scheduling systems exhibit the most significant decrease by 1.4 days. Further, implementation of Order Entry, Patient Tracking RFID, and EMAR applications, also reduce the LOS deviation between 0.9 and 1 day, on average.

Propensity score matching allows us to reliably estimate the effect of early discharges on future 30-day readmission risk. After matching patients with similar exogenous characteristics, we find that the 30-day readmission risk for patients discharged earlier than GMLOS (treated group) is 1.1% higher than patients who are discharged around the GMLOS (control group), as reported in Table 6. The results confirm that early discharge comes at the cost of higher readmission propensity, after accounting for patient heterogeneity.

5.4. Robustness Checks

We have shown that the deviation of LOS from standard GMLOS guidelines decreases after health IT implementation, and that further deviations in terms of early discharge increase higher readmission risk at the patient level. As a robustness check, we further investigate the relationship between the level of HIT and the average readmission rate at the hospital level. First, we select hospitals that maintain low readmission rates, and then examine the difference between hospitals that have fully implemented operational HIT applications and the ones that have not implemented any HIT, as reported in Table 7.

After conducting *t*-tests on the different levels of operational HIT status, we find that there are significant differences in the characteristics of the two groups. The *t*-tests show that, compared to hospitals with high IT implementation, hospitals that maintain low readmission rates with low levels of operational IT are distinguished by the following characteristics: they are more likely to exhibit a higher level of patient case mix, be located in an urban location, have higher average LOS, and exhibit greater deviation from the standard GMLOS. We find that, among hospitals with low readmission rates even without operational health IT, their patients tend to stay longer than the mandated GMLOS guidelines. For instance, hospitals with no operational HIT keep their patients almost 8 days longer, on average, than the GMLOS, while

Table 6 30-Day Readmission Risk after Propensity Score Matching

Variable	Sample	Treatment group	Control group	Difference	SE
Probability of 30-day readmission	Early Discharge[a]	0.098	0.087	0.011	0.004**

[a]Average Treatment Effect on the Early Discharge Group.

Table 7 Health IT and LOS Deviation among Low Readmission Hospitals

Level of operational IT	CMI	Urban	Teaching Hospitals	Avg. LoS	GMLOS − LOS
High	1.457	0.882	0.235	5.135	−0.799
None	1.556	1.000	0.625	11.712	−7.592
Pr > *F*	<0.0001	<0.0001	0.378	<0.0001	<0.0001

[a]diff. (GMLOS, LOS) = GMLOS − LOS.

hospitals with high levels of operational IT discharge their patients within a day of the GMLOS guideline. It is possible that hospitals which did not implement operational HIT may suffer in terms of the effectiveness of their care coordination processes, but they compensate by keeping patients long enough to provide appropriate care to reduce their readmission rates.

5.4.1. Health IT and Post-discharge Care Coordination. While operational IT-enabled care coordination may reduce hospital readmission risk and streamline adherence to standard procedures, other factors such as post-discharge care coordination, medication adherence after discharge, and the quality of care during hospital stay play important roles in determining patients' risk of readmission. Prior studies have suggested that care coordination and transitional care are keys to improve patient care outcomes (Coleman et al. 2006, Naylor et al. 1999, Peikes et al. 2009). For instance, Peikes et al. (2009) studied the effect of care coordination after discharge, such as nurses providing patient education and monitoring to improve adherence and ability to communicate with physicians. Based on randomized trials, they reported that hospitals that demonstrated improvements in patient health outcomes are more likely to provide patient educational resources, follow-up frequently with patients after discharge, and coordinate post-discharge care with patients' local clinics and primary care physicians. The study concluded that, unless hospitals' care coordination programs are accompanied by strong care transition initiatives, they are unlikely to improve healthcare outcomes or realize cost savings.

While these factors are unobservable in the context of our study, we conduct a limited scope of investigation to tease out the effect of care coordination on average readmission rates among the hospitals in our sample. Toward this end, we acquired and analyzed data from the American Hospital Association (AHA), which includes data on health IT use for care coordination in year 2009. Specifically, the AHA IT supplement annual survey includes questions on whether the hospital's "computerized system include patient summary of care record for relevant transitions in care." We find that the average readmission rate of hospitals that implemented a summary of care transition application is 2% lower than hospitals which did not implement or only had a partial implementation. However, we did not observe a reduction in the deviation of LOS among the same comparison groups. Therefore, we find limited evidence which shows that care coordination may reduce readmission risk, but cannot conclude unambiguously that it results in a corresponding decrease in the LOS deviation.

5.5. Cost Trade-offs: Readmission vs. Early Discharge

Our analysis suggests that current cost savings from early discharges might come at the expense of higher future costs due to increased risk of patient readmissions. To quantify this trade-off, we conduct additional analysis on "early discharge" patients. We classify early discharge visits into four groups based on the magnitude of their deviation from GMLOS. These groups represent patient visits where the patients were discharged one, two, three, four, or more days, earlier than GMLOS. For each group, we calculate the average cost of additional days of hospital stay.

Our claims dataset (from DFWHC) records total charges for each patient's hospitalization. The average per-day cost of a patient visit is then calculated as the total charges for this visit divided by LOS of the visit. Hence, we can infer the additional cost, had the hospital kept the patient longer, as the product of the average daily cost and the number of extra days needed to comply with the GMLOS guideline. We also observe the total charges related to each patient's readmission. Thus, we are able to calculate the total extra cost resulting from 30-day readmissions for all patients in each "early discharge" group. By comparing the expected additional cost for the extended hospital stay with the charges from readmissions, we can assess the cost trade-offs.

Our results, as reported in Table 8, indicate that the total cost savings associated with early discharges is significantly lower than the total cost of future readmissions. For example, while the total cost

Table 8 Total Cost–Benefit Trade-off Analysis

Frequency	Early discharge	Total 30-day readmission costs	Total extra cost
5294	Early 1 day	$228,769,669	$7,746,015
761	Early 2 days	$31,126,818	$11,857,010
422	Early 3 days	$15,514,579	$9,790,424
552	Early 4 days and over	$19,565,769	$18,593,518

of 30-day readmissions incurred by the 5,294 patients who are discharged 1 day earlier is more than \$228 million, the total cost of keeping those patients for one extra day (hypothetically) would amount to less than \$8 million. Likewise, the total expected cost of extra care for two extra days, for the 791 patients who are discharged 2 days earlier (than GMLOS), is calculated to be around \$12 million, which is still significantly lower than the \$31 million readmission expense. Hence, our analysis suggests that "early discharge" policies tend to be myopic, since they ignore the higher costs that are incurred in future readmissions.

6. Discussion

Our statistical analysis reveals that the deviation between the actual LOS and GMLOS has a significant impact on the risk of future 30-day readmissions. Specifically, our results suggest that patients with hospital stays shorter than the GMLOS by two or more days are likely to exhibit greater readmission risk, while patients whose LOS exceeds GMLOS are likely to exhibit a slightly lower risk of readmission. For instance, patients in the "Below GMLOS" group have a 7% higher readmission risk compared to patients in the control group where the LOS is within 1 day of the recommended GMLOS guidelines. Our analysis indicates that the cost savings from early discharges is outweighed by the extra costs incurred due to future readmissions. This suggests that a myopic policy of "saving a few dollars now' may backfire at the expense of 'spending more resources later." We also find that patient stickiness to the same hospital is associated with a reduction in readmission risk.

Further, we estimate the impact of health IT on LOS deviation, and how it pertains to the 30-day readmission risk. Our results indicate that implementation of HIT applications for operational coordination of patient care increases hospitals' adherence capabilities related to standard guidelines on length of stay. To the best of our knowledge, ours is one of the first studies examining the relationship between implementation of health IT and deviation of LOS from GMLOS. Our empirical findings reveal that there is a statistically significant association between the implementation of various hospital-level, health IT applications and reduction in the deviation between LOS and GMLOS guidelines. For instance, we observe that the average difference between actual LOS and standard GMLOS decreases significantly, after implementation of patient tracking RFID technology; and hospitals with patient tracking RFID applications have significantly lower variations in length of stay.

Our results have several implications for clinical practice, healthcare operations, and policy makers. Our findings suggest that clinicians and hospital operations managers should pay close attention to the deployment and effective usage of these HIT applications as a mechanism to reduce deviations from LOS guidelines. These HIT applications include clinical decision support, diagnostic and scheduling capabilities, that enable hospital decision makers to more effectively manage patient stays by reducing bottlenecks pertaining to congestion, admission and discharge delays, and improve patient satisfaction. For example, patient scheduling applications allow managers to optimally manage patient stays such that patients who require longer stays are not discharged preemptively; by the same token, patients who are at lower readmission risk can be discharged earlier, thereby freeing up available beds for high-risk patients.

Our research also suggests that patient characteristics, including gender, race, age, mortality risk, payer type, and discharge destination, all significantly impact the readmission risk. Specifically, female patients are 20.8% less likely than male patients to be readmitted, *ceteris paribus*; while African-American patients, as well as Medicaid patients are more likely to be readmitted compared to Caucasian and self-pay patients, respectively. This group of patients can benefit more from care coordination during their hospital stay as well as follow-up care.

From a societal perspective, our study has several policy implications with respect to the current debate on reductions in Medicare reimbursements for hospitals with high readmission rates (Helm et al. 2016). Our results suggest that, although shorter stays may result in lower short-term hospital costs, the long-term impact due to greater readmission risk exceeds the potential cost savings from early discharge. Hence, a singular focus on patient LOS, instead of its deviation from GMLOS, may lead to a myopic policy which would result in higher readmission costs due to greater frequency of patient readmissions (Joynt and Jha 2013). Furthermore, with the imposition of penalties by CMS due to preventable readmissions associated with CHF, hospitals may lose additional revenue due to reduced reimbursed rates for treating Medicare patients. Our study provides empirical evidence that discharging patients quicker, yet sicker, is not only a myopic action, but has long-term consequences on the overall cost of patient care.

7. Conclusions

In this research, we develop a new approach to jointly estimate hospital length of stay along with its impact on 30-day readmission risk, using a large panel of CHF patients. In order to account for the effect of CMS guidelines related to hospital length of stay, we propose a new performance measure, the *deviation* in LOS from CMS stipulated guidelines, to measure hospitals' operational performance related to length of stay. We conduct a longitudinal study of CHF patients in the North Texas region where we tracked their admissions across multiple hospitals over a period of 4 years. Our results indicate that (a) deviations of LOS from the stipulated GMLOS guidelines are significantly associated with readmission risk, and (b) health IT systems have a significant impact on hospitals' adherence to GMLOS guidelines. Our research represents one of the first studies to empirically examine and document the effect of health IT on hospital adherence to LOS guidelines, and its impact on subsequent patient readmission risk.

Nevertheless, our study is not immune to limitations. Our research is restricted to studying CHF patients within one geographic region. Although the North Texas region is fairly large and diverse in terms of its population, future studies are needed to account for patient demographic characteristics in other regions of the country. Future studies can extend these models to study other chronic diseases, such as diabetes and COPD, which are also characteristic of long hospital stays and high readmission rates. We rely on secondary, administrative claims data obtained from the region's hospitals, and are limited by the scope of this dataset. For example, administrative claims data do not include clinical data such as detailed clinical conditions of a patient or medications administered during a patient's hospital stay. Further, our data do not specify which IT applications are involved in different stages of the care delivery process at the level of individual patient visits. Therefore, we do not observe the direct consequences of HIT usage in patient care processes, which is another limitation due to the nature of secondary data.

Lastly, lack of detailed hospital operations data also limits our ability to control for organizational factors that may affect hospital operational efficiency and quality of care. For example, studies have found that organizational culture can affect the care quality such as a hospital's level of adherence to evidence-based standards and readmission rates (Senot et al. 2016). Organizational safety culture, such as the development and adherence to quality control policies (e.g., safety protocol and open and constructive responses to errors), is significantly associated with quality improvement in the healthcare industry (Hofmann and Mark 2006). While our use of the difference-in-difference approach and propensity score matching have largely alleviated the concerns of omitting time-invariant, hospital-level variables, organizational culture and quality control policies may gradually change over time. Investigating the inter-relationships between organizational culture, quality control policy, adherence to standards, and care quality would open an interesting avenue for future research.

Other future extensions to this research could also include obtaining clinical data that shed light on patient-specific, clinical details that may provide a deeper understanding of the causal associations between LOS and readmission risk. We also note that some patient visits in our data may include short stays where patients may leave the hospital early against medical advice. Hence, the LOS in such cases may not occur under the discretion of the hospital administration or the patient's care management team, and its impact on readmission risk cannot be attributed to the hospital's decision-making process.

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Notes

[1]Source: http://www.cms.gov/CCIIO/Resources/Fact-Sheets-and-FAQs/aca_implementation_faqs7.html

[2]GMLOS represents the national mean length of stay for each DRG, as determined and published by CMS. The use of a geometric mean, instead of a straight average LOS (ALOS), reduces the effect of very high or low values.

[3]A hospital's Case Mix Index (CMI) is a measure of the relative cost or resources needed to treat the mix of patients during a calendar year. It is calculated by summing the DRG weights assigned to each Medicare Severity-Diagnosis Related Groups (MS-DRG) released by CMS for all Medicare discharges, divided by the number of discharges.

[4]A limitation of the HIMSS data is that it does not provide the precise implementation date of the HIT applications. We acknowledge that a binary variable may not accurately reflect the level of HIT for an entire year of HIT implementation since it may occur in the middle of the year.

[5]Since hospitals may have implemented HIT applications at different times, our DID specification is effectively equivalent to a typical panel regression model.

[6]Due to space constraints, we do not present the comparison between the treatment and the control group with respect to these observable control variables.

[7]We also examined an alternative specification that includes hospital-specific fixed effects, where the estimate

of β_1 for our key variable remains qualitatively unchanged. We choose the specification in Equation (2) because these hospital-specific control variables are also of interest to us.
[8]Since we ran separate DID models for each HIT application, we only show the coefficients of the interaction term of the regression model due to space constraints.

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